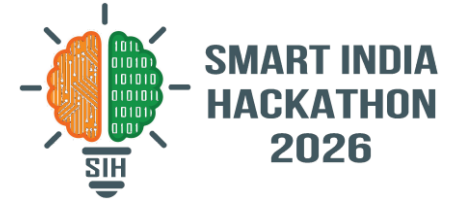


SMART INDIA HACKATHON 2026

TITLE PAGE



Problem Statement ID – SIH26165

Problem Statement Title – AI/NLP Engine to Detect Serious Injury & Fatality (SIF) Precursors in OIL's Unsafe-Act / Unsafe-Condition and Near-Miss Reports

Theme – Smart Automation

PS Category – Software

Team ID – _____

Team Name – Signal-0

SANKET
SIF Precursor Detection

Reads a free-text safety report and answers one question: could this have killed someone?

HAZARD



CONTROL STATUS



PLAUSIBLE SEVERITY

SANKET — EARLY SAFETY WARNINGS FROM INCIDENT NARRATIVES

Reads a safety or near-miss report and answers one question: could this have killed someone?

WHAT IT IDENTIFIES

- Hazard + Life-Saving Rule
- Control status
- Severity (1-5)
- SIF Precursor: Yes / No
- Evidence & reasoning

WHAT MAKES IT DIFFERENT

- Three gates, not one score
- One-Change severity rule
- LLM + local ML fallback
- Ranks a queue - never closes a report
- Every judgement logged

THE THREE GATES

GATE 1 · HAZARD

Is a high-energy hazard present?

yes · no · insufficient



GATE 2 · CONTROL

Was a barrier doing its job?

absent · failed · present · unclear



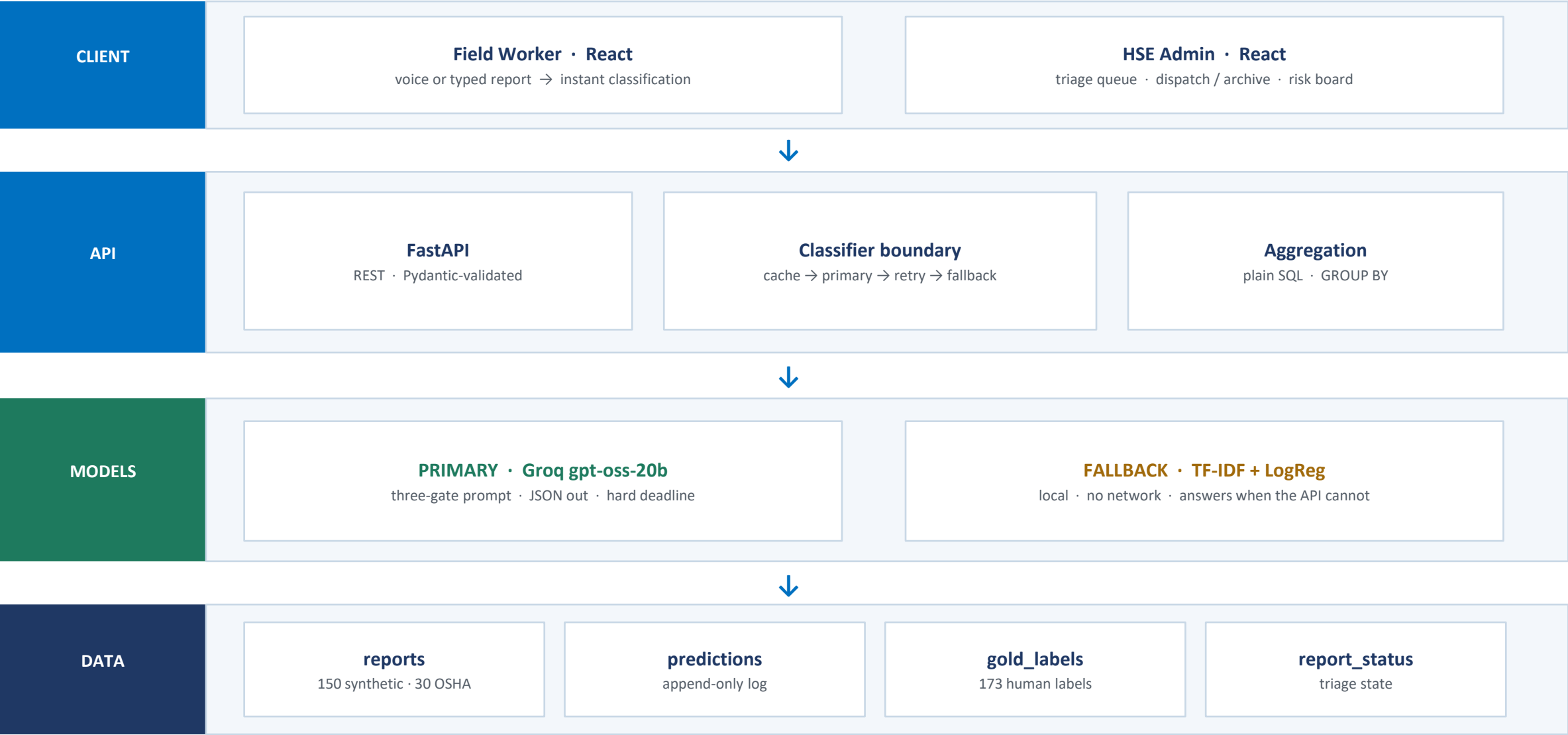
GATE 3 · SEVERITY

Plausible worst case?

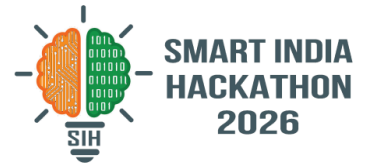
1 - 5

SIF PRECURSOR = Hazard YES + Control ABSENT / FAILED + Severity $\geq$ 4

TECHNICAL APPROACH



FEASIBILITY AND VIABILITY



FEASIBLE BECAUSE

- Uses incident reports that already exist
- No specialized hardware
- API-based & scalable
- LLM + local fallback
- Human-in-the-loop

CHALLENGE → SOLUTION

- Unstructured text → LLM analysis
- Ambiguous controls → 4-state classification
- API failure → retry + local fallback
- Domain variation → 30 real OSHA reports
- Model changes → one version per dashboard

OUR REAL LIMIT IS TOKENS, NOT COMPUTE

200,000

tokens / day, free tier

~3,150

tokens per report

63

reports / day

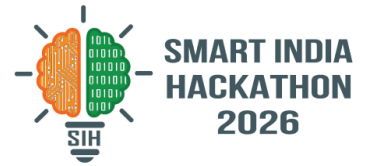
~2.8 days

full corpus re-run

SO THE PIPELINE IS BUILT TO SURVIVE IT

Resumable · never stores a fallback as if it were a real answer · dashboard always has data to show

IMPACT AND BENEFITS



POTENTIAL IMPACT

- Early detection of SIF precursors in OIL reports
- Prioritized review of high-risk incidents
- Ranks sites by precursor RATE, not count
- Finds recurring hazards across sites & shifts

BENEFITS

- Safety — act before the incident
- Operational — no manual first-pass screen
- Economic — prevents injuries and downtime
- Scalable — ten reports or ten thousand

Safety Reports → SIF Precursors → Risk Prioritization → Early Action → Safer Operations

WHY WE RANK BY RATE, NOT BY COUNT

A site that reports honestly logs more incidents. Counting them punishes it.

Count → the busiest site always looks worst

Rate → precursors ÷ reports, small groups held back

RESEARCH AND REFERENCES

RESEARCH

IOGP Life-Saving Rules · DEKRA SIF Framework · EEI SIF Model

DATA

150 synthetic reports + 30 real OSHA severe-injury narratives · 173 gold labels · 2 annotators

PROTOTYPE EVALUATION

HUMAN CEILING

0.947

Cohen's kappa

97.4% agreement · independent · n=38

TF-IDF BASELINE

0.754

F1 $\pm$ 0.077

5-fold CV · PR-AUC 0.847 · n=114

OSHA GENERALISATION

0.118

F1

real narratives · n=30, too small to conclude

LLM F1 withdrawn — we found a held-out report inside our own prompt and pulled the number rather than ship it. Re-measurement in progress.

STATED UP FRONT, BECAUSE A JUDGE WILL FIND THEM

Precursor rate 58.7%

not the 20–25% cited — our generator centred every report on a hazard

Ceiling is n=38

a re-label subset, so we quote the subset, never a bare kappa

No accuracy figure

and never a baseline scored on its own training data

GitHub: SANKET — AI-powered SIF precursor detection